Differentiating No-Action Outcomes in Agent Evaluations

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Motivation

Zhu et al. [1] showed that τ bench gives credit to agents that do nothing. Testing now on τ^2 bench, a null agent that takes no action and says nothing, gets 46% on airline, passing 23 of 50 tasks, and 5.3% on retail, passing 6 of 114 tasks.

This occurs because for these tasks, taking no action is the correct response from an agent. Zeroing in on the airline tasks, these 23 tasks test whether the agent follows its given policy when the user pushes it to break the policy. This is a known aspect of airline-related tasks that an agent should be tested on. For example, a normal airline customer can call in asking to change the destination of a ticket they already bought, even though it might be against company policy to make the requested change [2].

However, the problem is that an agent should understand that no action is the correct course of action *for the right reasons*. We argue that accidentally stumbling into the correct course of action for no-action tasks is common and likely enough to meaningfully affect the efficacy of the final accuracy and especially the relative rankings of different models.

To show this, we ran an experiment where we tested 5 real, distinct models on the subset of 23 no action airline tasks and then combed through each trajectory with Opus 5.5 to hand-grade whether the verdict was given for the "correct reasons". We define correct reasons with a rubric that checks if

1. The agent ran the necessary read tools to learn the evidence it needed to reason
2. The agent tried or agreed to run forbidden write tools
3. The agent prematurely transfers responsibility to a human before expected reasoning

With this hand-grading, we got the following results:

Model	Evaluator Accuracy	Hand-Graded True Accuracy	Passes for the Wrong Reason	Wrong-Reason Pattern
claude-sonnet-4.5	21	17	4	mixed
gpt-4.1	21	12	9	agreed to a forbidden write (6)
gpt-4.1-mini	20	11	9	agreed to a forbidden write (5)
llama-3.3-70b	21	0	21	unneeded transfer (17)
gpt-4.1-nano	4	0	4	19 harmful writes (evaluator caught)

This is an issue because some models like llama-3.3-70b get 42% accuracy unjustly padded to their evaluation while other models like claude-sonnet-4.5 only get 8% accuracy unjustly padded to their evaluation. This discrepancy in padding can easily lead to incorrect relative rankings or comparisons.

Therefore, we aim to differentiate different no action outcomes from agents to find an effective method to classify where a no action response was given due to proper reasoning.

Goals

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Minimal	Modify τ^2 bench such that no action verdicts are differentiated as justified or unjustified.
Target	Look at 1-2 other benchmarks and try to fix similar outcome validity issues.
Stretch	Create a generalizable framework that targets this validity issue for any benchmark.

References

[1] Y. Zhu et al. *Establishing Best Practices for Building Rigorous Agentic Benchmarks.* arXiv 2507.02825, 2025.

[2] V. Barres et al. *τ^2 Bench: Evaluating Conversational Agents in a Dual Control Environment.* arXiv 2506.07982, 2025.